Noninvasive Measurements of Pyridine Nucleotide and Flavoprotein in the Lens

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Abnormalities in glucose metabolism are thought to be among the main causes of cataract formation. The authors have made noninvasive biochemical measurements of the lens that provide information concerning glucose metabolism in the lens epithelium. The autofluorescence of reduced pyridine nucleotides (PN) and oxidized flavoproteins (Fp) within the rabbit lens were noninvasively measured as a function of depth using redox fluorometry. The peak of the autofluorescence at 440 nm (excited at 360 nm) and 540 nm (excited at 460 nm) were determined at the lens epithelium. When 8 mM sodium pentobarbital, a known inhibitor of mitochondrial respiration, was applied to the lens, the autofluorescence peak at 440 nm increased and that at 540 nm decreased. The 440 nm autofluorescence is thought to be from reduced pyridine nucleotides, whereas the 540 nm autofluorescence is from the oxidized flavoprotein. Blocking lens respiration with pentobarbital caused an increase in the PN/Fp ratio by a factor of 3 within 3.5 hr after pentobarbital application. Invest Ophthalmol Vis Sci 28:785–789, 1987

Abnormalities in glucose metabolism are thought to be among the important causes of cataract formation. The metabolic pathways of oxidative phosphorylation, the citric acid cycle, the hexose monophosphate shunt, glycolysis, and the sorbitol pathway, are active in the lens.[1] These pathways are under delicate controls and interact with one another. The calf lens is able to maintain its energy-requiring processes equally well anaerobically as aerobically as long as glucose is present. Thus the anaerobic pathway is thought to play a prominent role to produce enough adenosine triphosphate (ATP) to maintain a normal metabolic state in an environment of low-oxygen tension such as lens.[2] Many studies of glucose metabolism in the lens have been reported,[3-10] but few used noninvasive methods. In order to obtain noninvasive measurement of lens metabolism, we applied ocular redox fluorometry[11,12] to the rabbit lens.

Noninvasive redox fluorometry measures the fluorescence intensity of the reduced pyridine nucleotides (NADH and NADPH) in mitochondria and cytoplasm and the oxidized flavoproteins (Fp) in mitochondria.[11-17] These redox signals reflect the oxidation–reduction (redox) state of the cells being measured. The redox state is dependent on oxygen supply and use, substrate availability, and the metabolic work load of the tissue. Because NAD and NADH are predominantly involved in glycolysis and the reduction of pyruvate to lactate, redox fluorometry has a potential value to determine the functional state of the lens.

Redox fluorometry was developed by Chance[18] to measure the metabolic state of mitochondria in suspension. Ocular redox fluorometry, originally developed by Laing et al,[11] has been applied in a variety of ocular studies.[12-17] In this article we report how ocular redox fluorometry was used to provide information concerning the biochemical state of the lens epithelium and anterior cortex as a function of depth within the lens.

Materials and Methods

Instrumentation

Redox fluorophotometry in combination with specular microscopy was used to measure the PN/Fp ratio in the lens. A block diagram of the instrumentation used is shown in Figure 1. The basic instrument used has been previously described.[11,12] The current system is controlled by a microcomputer and provides automatic depth scanning, automatic filter changing, data storage, and analysis. Using this instrumentation, three signals were measured depending on the filter combinations that were used. One fluorescence signal was

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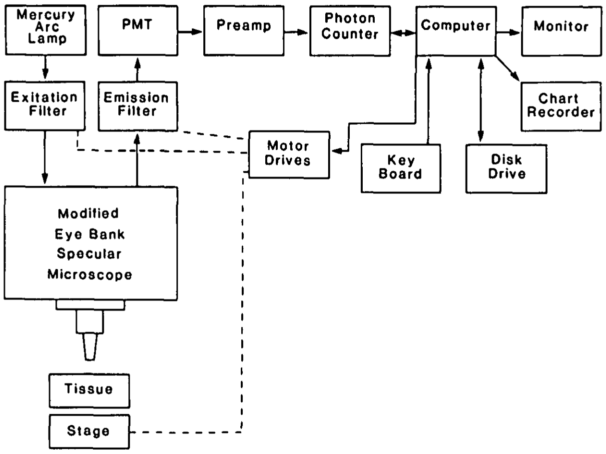

Fig. 1. A block diagram of the instrument used to measure the reduced pyridine nucleotides and the oxidized flavoprotein redox signals in the rabbit lens.

measured by exciting with 366 nm wavelength and measuring wavelengths centered about 440 nm (the 366/440 signal). A second fluorescence signal was measured by exciting at 460 nm and measuring wavelengths centered about 540 nm (the 460/540 signal). A reflection signal was measured by exciting at 540 nm and measuring at 540 nm, and was used as a reference for localizing the epithelium peak.

Preparation of the Lens

New Zealand albino rabbits, weighing 3 kg, were anesthetized with an intramuscular injection of 0.5–1.0 ml of 1:1 ketamine HCl (30 mg/kg of body weight) and xylazine (6 mg/kg of body weight). Retrobulbar anesthesia using 0.5 ml of 4% xylocaine was performed and eyes were enucleated. An overdose of pentobarbital sodium into vein was then used to kill the animal in order to prevent the respiratory inhibiting effect of pentobarbital. A trephine for penetrating keratoplasty of 3-mm diameter was applied to the center of cornea, and the corneal button was removed carefully without any damage to the iris or lens. The globe was then put into the chamber filled with buffered salt solution (BSS+) with the cornea side up. Rabbits were cared for and treated in accordance with the ARVO Resolution on the Use of Animals in Research.

Measurements

For the measurements, the glass dipping cone of the ×20 objective lens (Bio-Optics, Arlington, MA) was inserted through the hole in the cornea to within 1 mm from the epithelial lens surface. To improve the depth resolution of the instrument, the slit width was decreased to 90 μm, which gives us about 4.5 (90/20) μm wide image on the surface of the lens. The depth resolution of the instrument was determined by measuring the reflection signal as a function of depth by scanning across the surface of a glass microscope slide. The depth resolution was taken as the half-width of this signal at half maximum signal height. For the 90 μm slit used, this depth resolution was 50 μm. With this setting, the epithelium measurement actually includes epithelium and cortex. At the high magnification used, the measurement could only be made in the anterior part of the lens because of the 2 mm maximum working distance of ×20 objective lens. To change the metabolic state of the lens from normal, the BSS+ bathing the globe was replaced by BSS+ containing 8 mM sodium pentobarbital, a known inhibitor of mitochondrial respiration. The lens autofluorescence as a function of depth was measured at room temperature at various times before and after the application of the drug.

Results

In the normal fresh rabbit lens, fluorescent emission signals were observed by exciting the lens at 366 nm and measuring at 440 nm (the 366/440 signal), and by exciting at 460 nm and measuring at 540 nm (the 460/540 signal). Figure 2 shows the 366/440 signal and the 460/540 signal for a depth scan in anterior part of a fresh lens. The maximum signal peak was always seen at the lens capsule/epithelial position. The lens cortex signal was nearly constant with an average value smaller than epithelium peak. Representative values for these two signals expressed in arbitrary units are 1,500 and 800 for the 366/440 signal (noise level was 250), and 220 and 60 for the 460/540 signal (noise level was 60). The 440/540 signal in the cortex was always within the noise level. Figure 3 shows the changes in the two signals with time after application of the 8 mM pentobarbital. For the duration of the experiment (3.5 hr), the 366/440 signal kept increasing up to 2,350, and the 460/540 signal kept decreasing down to 115 in the capsule/epithelium complex. The signal levels in the cortex also increased to 1,600 for the 366/540 signal but did not change for the 460/540 signal. Figure 4 shows the time dependence of the ratio of these two signals. In the epithelium/capsule complex the ratio just before the addition of the respiratory inhibitor was 6.8. The ratio increased to 20.4 after 3.5 hr. In the cortex the ratio was 13.3 just before application and increased to 22.9 by 3.5 hr. During the measurement, neither the transparency of the lens or the reflection signal changed.

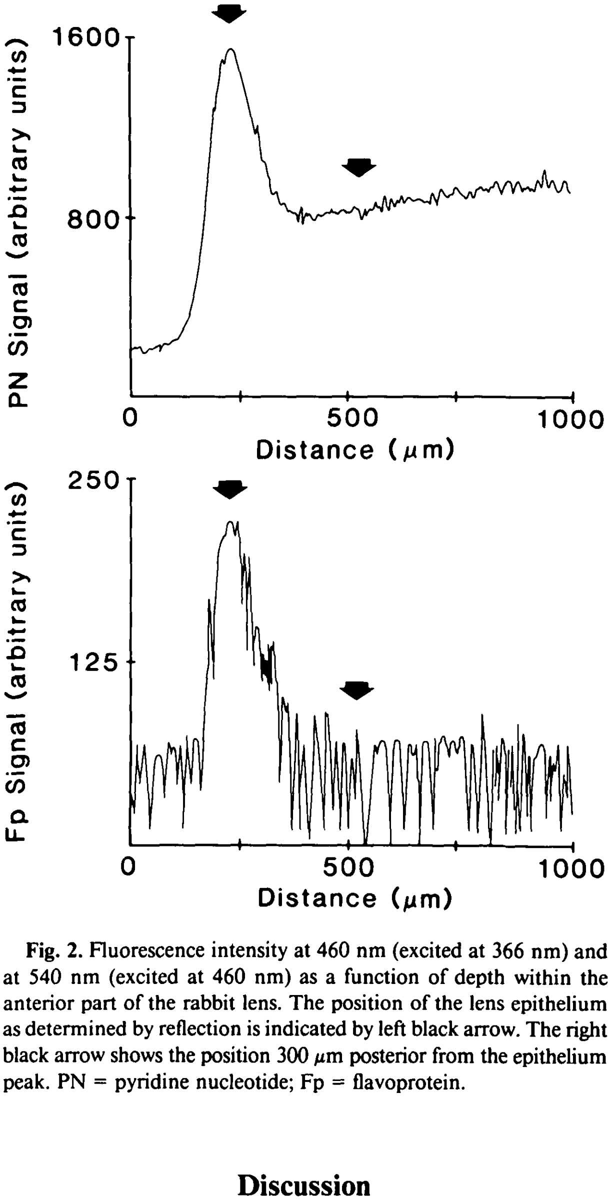

Fig. 2. Fluorescence intensity at 460 nm (excited at 366 nm) and at 540 nm (excited at 460 nm) as a function of depth within the anterior part of the rabbit lens. The position of the lens epithelium as determined by reflection is indicated by left black arrow. The right black arrow shows the position 300 μm posterior from the epithelium peak. PN = pyridine nucleotide; Fp = flavoprotein.

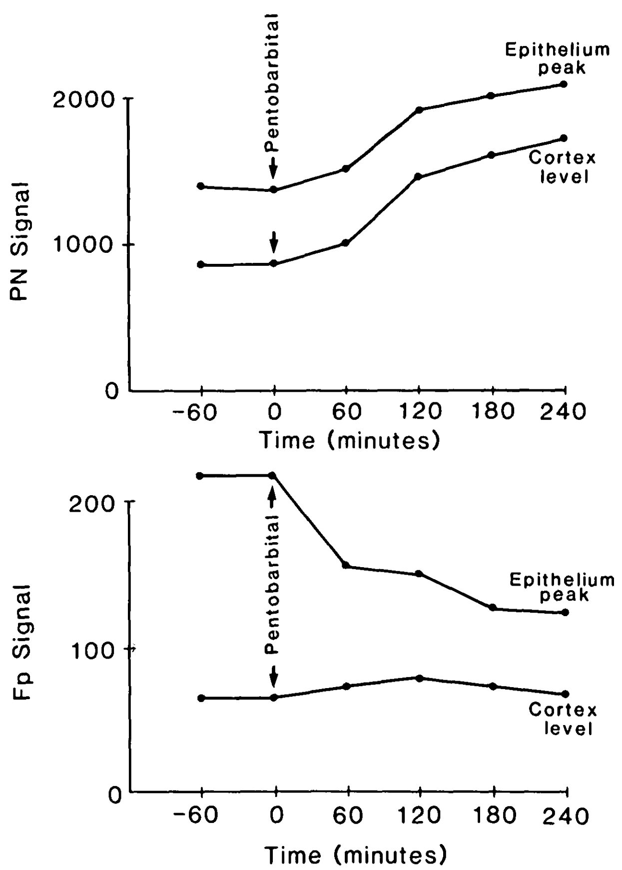

Fig. 3. Changes in the pyridine nucleotide (PN) and flavoprotein (Fp) fluorescence intensity from lens epithelium and cortex before and after the application of 8 mM sodium pentobarbital. The epithelium measurement and cortex measurement were made at arrows in Figure 2. The PN signal increased, and the Fp signal decreased.

Discussion

In the lens, there are two major types of fluorochromes. One type can be excited at 290 nm and emits around 340 nm (the purple fluorescence), and the other type can be excited at 340 nm and emits at 420 nm (the blue fluorescence).[19] The purple fluorescence is the major fluorescent component of the lens, is usually attributed to a protein-bound tryptophan, and is relatively constant during the aging process.[19,20] The blue fluorescence has been observed to be protein linked, appears to increase with age, and is thought to result from anthranilic acid and bityrosine and possibly other nontryptophan derivatives.[21] Among this group, β-carboline also has been identified as a major fluorochrome in aging and cataractous human lenses.[22] In addition to these two types of fluorochromes, a red

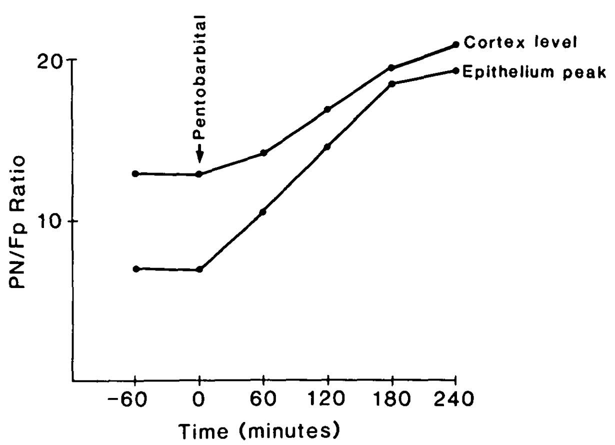

Fig. 4. Changes of PN/Fp ratio in lens epithelium and cortex before and after the application of 8 mM sodium pentobarbital. The ratio of epithelium increased by a factor of 3, and the ratio of cortex increased by a factor of 1.8.

fluorescence, which has an emission maximum at 672 nm with excitation at 647 nm, has been found recently in older human lenses.[23]

In making redox measurements, the reduced PN is measured by exciting at 366 nm and measuring at 440 nm, and the oxidized Fp are measured by exciting at 460 nm and measuring at 540 nm. Of the three lens fluorochrome types, the purple and red fluorochromes do not effect our 366/440 and 460/540 signals because the excitation and emission wavelengths of the purple and red fluorochromes are sufficiently different from those used by us to measure our two fluorescence signals. However, the age-dependent blue fluorescence does interfere with the PN fluorescence so that our 366/440 signal could contain two components: a blue fluorescence signal and a PN signal. The fact that our measurements changed dramatically after application of pentobarbital, a respiration inhibitor, implies that the changes we measured came from the fluorescence of pyridine nucleotides because neither tryptophan nor the other fluorescent amino acids or the other chromphores would be affected by the application of a respiration inhibitor during such a short time. So even though the fluorescence signals do not arise exclusively from the PN, the contributing fluorescence of PN are enough to measure, and redox fluorometry can be applied to the study of lens respiration.

Because both glucose and oxygen are primarily used by the lens epithelium, and the ATP and enzyme levels in the lens are the highest in this area,[24] it is not surprising that the PN and Fp signals have a maximum at the lens epithelium. In contrast to the presence of PN signals in the cortex, we found no measurable Fp signal in the cortex. This is to be expected because the Fp signal, which arises only from the mitochondrial Fp,[25] should be small because there are very few, if any, mitochondria in the lens cortex. Because the PN signal comes from the reduced form of NADH and the Fp signal comes from the oxidized form of Fp, these parameters ordinarily move in opposite directions in response to the same stimulus as they did in these experiments. Stewart and Augusteyn[26] determined the oxidized nicotinamide adenine dinucleotide (NAD^+), NADH, oxidized nicotinamide adenine dinucleotide phosphate ($NADP^+$), and NADPH concentrations in rabbit lens cortex using the enzymatic cycling method and found that more than 99% of the PNs are NADH and NAD+, and less than 1% consists of $NADP^+$ and NADPH. This means that essentially all of the PN signal that we measured is because of NADH.

Perhaps more important than the steady-state concentration of the reduced pyridine nucleotides is the capacity of the lens to generate a reducing potential. It is our major concern that NAD^+ and NADH are predominantly involved in glycolysis and the reduction of pyruvate to lactate.[1] In addition to this NADH synthesis, a recent study showed that the sorbitol pathway is activated in lenses incubated in high glucose or galactose and in the diabetic rat lens with conversion of aldose to sugar alcohol, resulting in NADH production.[27,28] This means that the sorbitol pathway, glycolysis and synthesis of α-glycerophosphate are linked via the redox state of NAD^+/NADH in the lens. Thus there is a large possibility that metabolic changes of diabetic lens can be monitored through lens redox fluorometry.

Because there are few mitochondria in the lens, and anaerobic glycolysis occurring in the cytoplasm is the major glycolytic pathway, our PN measurements are not considered to come from the mitochondria but chiefly refer to anaerobic glycolysis and the α-glycerophosphate cycle. Because the Fp signal comes from only mitochondrial Fp, this signal can be an indicator of mitochondrial function of lens epithelium. Thus this technique has great value in measuring changes in anaerobic glycolysis, epithelial mitochondrial function, and sorbitol pathway noninvasively through the changes of relative concentration of NADH and Fp.

Key words: redox fluorometry, lens metabolism, glycolysis, pyridine nucleotide, flavoprotein

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